

Kramers-restricted relativistic pCCD: amplitude equations, Λ equations and density matrices

Generalisation of Henderson, Bulik, Stein, Scuseria, J. Chem. Phys. **141**, 244104 (2014) (henceforth HBSS) to a Kramers-restricted spinor basis including spin–orbit coupling (2c, X2C or 4c no-pair). Orbital gradient and Hessian are not given; only the quantities entering them (γ and Γ).

1 Conventions

Kramers basis. Spinors come in Kramers pairs $\{\phi_p, \phi_{\bar{p}}\}$ with

$$\phi_{\bar{p}} = \mathcal{K}\phi_p, \quad \mathcal{K}\phi_{\bar{p}} = -\phi_p, \quad \mathcal{K}^2 = -1, \quad (1)$$

e.g. $\mathcal{K} = -i\sigma_y\mathcal{C}_0$ (2c) or $-i(\sigma_y \oplus \sigma_y)\mathcal{C}_0$ (4c). Lower-case indices label Kramers *pairs*: i, j occupied, a, b virtual, p, q, r, s general. Upper-case P, Q, R, S label individual spinors ($P \in \{p, \bar{p}\}$). The number of pairs is n , the number of electrons $N = 2n_{\text{occ}}$.

Pair operators and seniority.

$$P_p^\dagger = a_p^\dagger a_{\bar{p}}^\dagger, \quad P_p = (P_p^\dagger)^\dagger = a_{\bar{p}} a_p, \quad N_p = P_p^\dagger P_p, \quad (2)$$

and seniority is $\Omega = \hat{N} - 2\sum_p \hat{n}_p \hat{n}_{\bar{p}}$. On the seniority-zero space $N_p \in \{0, 1\}$ is the *pair* occupation and $\hat{n}_p = \hat{n}_{\bar{p}} = N_p$. The pair operators are hard-core bosons, $[P_p, P_q^\dagger] = \delta_{pq}(1 - 2N_p)$ on this space (the same algebra as for $\uparrow\downarrow$ pairs).

Note that $P_p^\dagger$ is invariant under the phase freedom $\phi_p \rightarrow e^{i\theta}\phi_p$ (which implies $\phi_{\bar{p}} \rightarrow e^{-i\theta}\phi_{\bar{p}}$) and, more generally, under any $SU(2)$ rotation inside the Kramers pair. All pair quantities below are therefore invariant under the Kramers-restricted (quaternion) gauge freedom, *provided* $\phi_{\bar{p}} = \mathcal{K}\phi_p$ is enforced exactly (it is automatic for quaternion-structured MO coefficients; it is *not* automatic in a GHF/DHF code that does not impose Kramers symmetry).

Hamiltonian and integrals.

$$H = \sum_{PQ} h_{PQ} a_P^\dagger a_Q + \frac{1}{2} \sum_{PQRS} \langle PQ|RS \rangle a_P^\dagger a_Q^\dagger a_S a_R, \quad \langle PQ|RS \rangle = (PR|QS), \quad (3)$$

$\langle PQ||RS \rangle = \langle PQ|RS \rangle - \langle PQ|SR \rangle$. For a time-reversal-even Hamiltonian, $\langle \mathcal{K}A|h|\mathcal{K}B \rangle = h_{BA}$ and $\langle \mathcal{K}A\mathcal{K}B|\mathcal{K}C\mathcal{K}D \rangle = \langle AB|CD \rangle^* = \langle CD|AB \rangle$; in particular $h_{\bar{p}\bar{p}} = h_{pp} \in \mathbb{R}$. Coulomb, exchange and “opposite-Kramers” exchange integrals are

$$J_{pq} = (pp|qq), \quad K_{pq} = (pq|qp) = \iint \frac{\rho_{pq}(1)\rho_{pq}^*(2)}{r_{12}}, \quad K_{p\bar{q}} = (p\bar{q}|\bar{q}p) = \iint \frac{\rho_{p\bar{q}}(1)\rho_{p\bar{q}}^*(2)}{r_{12}}, \quad (4)$$

with $\rho_{AB} = \phi_A^\dagger \phi_B$. For a spin-free scalar kernel (Coulomb), using $\rho_{\bar{p}\bar{q}} = \rho_{qp}$ and $\rho_{p\bar{q}} = -\rho_{q\bar{p}}$ (hence $\rho_{p\bar{p}} \equiv 0$ pointwise):

$$J_{p\bar{q}} = J_{pq}, \quad K_{p\bar{p}} = 0, \quad K_{pq} \geq 0, \quad K_{p\bar{q}} \geq 0. \quad (5)$$

In the nonrelativistic limit ($\phi_p = \varphi_p\alpha$, $\phi_{\bar{p}} = \varphi_p\beta$) $K_{p\bar{q}} = 0$; with spin–orbit coupling $K_{p\bar{q}} \neq 0$ — this is the opposite-spin exchange.

2 Seniority-zero Hamiltonian

Because $|0\rangle$, T and Z are all seniority-zero, every quantity in pCCD (E , residuals, Λ equations, γ , Γ) involves only the block of H between seniority-zero states. Acting on that space, the only surviving terms of H are (i) diagonal one-body terms, (ii) density–density terms between spinors, and (iii) pair transfer $P_p^\dagger P_q$ (terms such as $a_p^\dagger a_q^\dagger a_{\bar{q}} a_{\bar{p}}$ or $a_p^\dagger a_{\bar{p}}$ annihilate every seniority-zero state). One obtains

$$H_{\Omega=0} = \sum_p \varepsilon_p N_p + \sum_{p \neq q} G_{pq} P_p^\dagger P_q + \frac{1}{2} \sum_{p \neq q} W_{pq} N_p N_q \quad (6)$$

with, for a general time-reversal-even interaction (Coulomb, Gaunt, Breit),

$$\varepsilon_p = h_{pp} + h_{\bar{p}\bar{p}} + \langle p\bar{p} | p\bar{p} \rangle = 2h_{pp} + V_p, \quad V_p \equiv \langle p\bar{p} | p\bar{p} \rangle, \quad (7)$$

$$G_{pq} = \langle p\bar{p} | q\bar{q} \rangle = \langle p\bar{p} | q\bar{q} \rangle - \langle p\bar{p} | \bar{q}q \rangle = (pq | \bar{p}\bar{q}) - (p\bar{q} | \bar{p}q), \quad (8)$$

$$W_{pq} = \sum_{P \in \{p, \bar{p}\}} \sum_{Q \in \{q, \bar{q}\}} \langle PQ | PQ \rangle = 2(\langle pq | pq \rangle + \langle p\bar{q} | p\bar{q} \rangle). \quad (9)$$

Time-reversal symmetry makes G and W *real symmetric* matrices ($\langle p\bar{p} | q\bar{q} \rangle$ and $\langle p\bar{p} | \bar{q}q \rangle$ are each equal to their complex conjugates). Hence the pCCD amplitudes and all pair-level density matrices below are real, even though the spinors, h_{PQ} and the general integrals are complex. Complex/quaternion algebra only enters through the integral transformation and the orbital rotations.

Coulomb (Dirac–Coulomb, X2C–Coulomb, 2c–ECP) interaction. Using Eq. (5),

$$\varepsilon_p = 2h_{pp} + J_{pp}, \quad G_{pq} = K_{pq} + K_{p\bar{q}}, \quad W_{pq} = 4J_{pq} - 2K_{pq} - 2K_{p\bar{q}} \quad (10)$$

(the relative sign in G comes from $\rho_{\bar{p}q} = -\rho_{p\bar{q}}^*$, giving $(p\bar{q} | \bar{p}q) = -K_{p\bar{q}}$). For Gaunt/Breit terms (current densities are time-odd) Eqs. (5) do not hold and the general forms (7)–(9) must be used; in particular $V_p \neq J_{pp}$ in general.

Dictionary to HBSS.

HBSS (spin-orbitals, spatial indices)	nonrelativistic value	Kramers-restricted
$v_{qq}^{pp}, p \neq q$	K_{pq}	G_{pq}
v_{pp}^{pp}	J_{pp}	V_p
$2(v_{pq}^{pq} - v_{qp}^{pq}), p \neq q$	$4J_{pq} - 2K_{pq}$	W_{pq}
f_p^p	$h_{pp} + \sum_j (2J_{pj} - K_{pj})$	f_{pp} of Eq. (12)

With this substitution Eqs. (12), (19), (A8)–(A11) of HBSS hold verbatim; below they are written directly in terms of ε, G, W , which avoids the bookkeeping of the $b = a$ and $j = i$ ladder terms.

Reference energy and Fock diagonal.

$$E_{\text{ref}} = \sum_i \varepsilon_i + \frac{1}{2} \sum_{i \neq j} W_{ij} = \sum_I h_{II} + \frac{1}{2} \sum_{IJ} \langle IJ | IJ \rangle \\ \stackrel{\text{Coul.}}{=} \sum_i 2h_{ii} + \sum_{ij} (2J_{ij} - K_{ij} - K_{i\bar{j}}), \quad (11)$$

$$f_{pp} = h_{pp} + \sum_{J \in \text{occ}} \langle pJ | pJ \rangle = h_{pp} + \sum_j (\langle pj | pj \rangle + \langle p\bar{j} | p\bar{j} \rangle) \stackrel{\text{Coul.}}{=} h_{pp} + \sum_j (2J_{pj} - K_{pj} - K_{p\bar{j}}). \quad (12)$$

The Hartree, same-Kramers exchange and opposite-Kramers exchange contributions are explicit. (In (11) the $i = j$ Coulomb term reduces to J_{ii} since $K_{ii} = J_{ii}$, $K_{i\bar{i}} = 0$.)

3 pCCD energy and amplitude equations

$|\Psi\rangle = e^T|0\rangle$, $T = \sum_{ia} t_i^a P_a^\dagger P_i$, $\bar{H} = e^{-T} H e^T$, $E = \langle 0|\bar{H}|0\rangle$, $0 = R_{ia} \equiv \langle 0|P_i^\dagger P_a \bar{H}|0\rangle$.

Energy.

$$E = E_{\text{ref}} + \sum_{ia} G_{ia} t_i^a. \quad (13)$$

Amplitude equations. Define

$$\Delta_{ia} = \varepsilon_a - \varepsilon_i + \sum_{j \neq i} (W_{aj} - W_{ij}) = 2(f_{aa} - f_{ii}) + V_a + V_i - W_{ia}, \quad (14)$$

$$\Sigma_a^{(v)} = \sum_j G_{ja} t_j^a, \quad \Sigma_i^{(o)} = \sum_b G_{ib} t_i^b, \quad Y_{ji} = \sum_b G_{jb} t_i^b. \quad (15)$$

Then

$$\boxed{0 = R_{ia} = G_{ia} + \left[\Delta_{ia} - 2(\Sigma_a^{(v)} + \Sigma_i^{(o)}) + 2G_{ia} t_i^a \right] t_i^a + \sum_{b \neq a} G_{ab} t_i^b + \sum_{j \neq i} G_{ji} t_j^a + \sum_j t_j^a Y_{ji}} \quad (16)$$

All terms cost $\mathcal{O}(n_{\text{occ}}^2 n_{\text{vir}} + n_{\text{occ}} n_{\text{vir}}^2)$. A quasi-Newton update $t_i^a \leftarrow t_i^a - R_{ia}/\Delta_{ia}$ (with DIIS) is the usual choice.

For two pairs (one i , one a) this reduces to $G + (\varepsilon_a - \varepsilon_i)t - Gt^2 = 0$, i.e. exact diagonalisation of the 2×2 pair Hamiltonian.

4 Λ (z -amplitude) equations

$\langle L| = \langle 0|(1 + Z)$, $Z = \sum_{ia} z_a^i P_i^\dagger P_a$, $\mathcal{L}(t, z) = E(t) + \sum_{ia} z_a^i R_{ia}(t)$, and $\partial \mathcal{L}/\partial t_i^a = 0$ gives

$$\boxed{0 = G_{ia} + \left[\Delta_{ia} - 2(\Sigma_a^{(v)} + \Sigma_i^{(o)}) + 4G_{ia} t_i^a \right] z_a^i - 2G_{ia} \left(\sum_j z_a^j t_j^a + \sum_b z_b^i t_i^b \right) + \sum_{b \neq a} G_{ab} z_b^i + \sum_{j \neq i} G_{ij} z_a^j + \sum_j z_a^j Y_{ij} + \sum_b z_b^i X_{ab}, \quad X_{ab} = \sum_j G_{ja} t_j^b.} \quad (17)$$

Linear in z ; same diagonal preconditioner Δ_{ia} as the t equations. In practice $z_a^i \approx t_i^a$ is a good starting guess.

5 Density matrices

All expectation values are biorthogonal: $\langle O \rangle = \langle 0|(1 + Z)e^{-T} O e^T|0\rangle$.

5.1 Pair-level (independent) quantities

Intermediates (HBSS Eqs. (A9), (A11)):

$$x_i^j = \sum_a t_i^a z_a^j, \quad x_a^b = \sum_i t_i^b z_a^i, \quad x_i^a = \sum_{jb} t_i^b t_j^a z_b^j, \quad x_i \equiv x_i^i. \quad (18)$$

Occupation numbers ($n_p = \langle N_p \rangle$, identical for p and $\bar{p}$):

$$\boxed{n_i = 1 - x_i = 1 - \sum_a t_i^a z_a^i, \quad n_a = x_a^a = \sum_i t_i^a z_a^i, \quad 2 \sum_p n_p = N.} \quad (19)$$

Pair-transfer matrix $D_{pq} = \langle P_p^\dagger P_q \rangle$ (moves a pair $q \rightarrow p$), $D_{pp} = n_p$:

$$\boxed{\begin{aligned} D_{ij} &= x_i^j = \sum_a t_i^a z_a^j \quad (i \neq j), & D_{ab} &= x_a^b = \sum_i t_i^b z_a^i \quad (a \neq b), \\ D_{ai} &= z_a^i, & D_{ia} &= t_i^a + x_i^a - 2t_i^a(n_a + x_i - t_i^a z_a^i). \end{aligned}} \quad (20)$$

D is *not* symmetric ($D_{ia} \neq D_{ai}$); use it as is in a Lagrangian gradient, or symmetrise ($\frac{1}{2}(D + D^T)$) only if the gradient code assumes a symmetric 2-RDM (the energy is unaffected because G is symmetric).

Pair density–density matrix $Q_{pq} = \langle N_p N_q \rangle$, $p \neq q$:

$$\boxed{Q_{ij} = 1 - x_i - x_j \quad (i \neq j), \quad Q_{ia} = Q_{ai} = n_a - t_i^a z_a^i, \quad Q_{ab} = 0 \quad (a \neq b).} \quad (21)$$

($Q_{ab} = 0$ is exact for linear Z : the bra cannot de-excite two pairs.)

Energy in terms of pair RDMs (valid for any t, z ; equals $\mathcal{L}$):

$$\mathcal{L} = \sum_p \varepsilon_p n_p + \sum_{p \neq q} G_{pq} D_{pq} + \frac{1}{2} \sum_{p \neq q} W_{pq} Q_{pq}. \quad (22)$$

5.2 Spinor-level 1-RDM

Convention of HBSS: $\gamma_Q^P = \langle a_Q^\dagger a_P \rangle$. In the Kramers basis defining the pairs,

$$\gamma_q^p = \gamma_{\bar{q}}^{\bar{p}} = \delta_{pq} n_p, \quad \gamma_{\bar{q}}^p = \gamma_q^{\bar{p}} = 0. \quad (23)$$

The pairing spinors are the natural spinors; the Kramers-summed occupations are $2n_p$ (HBSS Eq. (A8)).

5.3 Spinor-level 2-RDM

Normalisation. We use the Löwdin normalisation

$$\Gamma_{RS}^{PQ} = \frac{1}{2} \langle a_R^\dagger a_S^\dagger a_Q a_P \rangle, \quad \sum_{PQ} \Gamma_{PQ}^{PQ} = \frac{N(N-1)}{2}, \quad (24)$$

so that

$$E = \sum_{PQ} h_{PQ} \gamma_P^Q + \sum_{PQRS} \langle PQ|RS \rangle \Gamma_{PQ}^{RS} = \sum_{PQ} h_{PQ} \gamma_P^Q + \frac{1}{2} \sum_{PQRS} \langle PQ||RS \rangle \Gamma_{PQ}^{RS}. \quad (25)$$

This is half the HBSS Γ (their Eq. (26b) has no factor $\frac{1}{2}$, and their energy/gradient, Eq. (25), carries the corresponding $\frac{1}{2}$). If the existing gradient/Hessian was coded with the HBSS normalisation, multiply its two-body terms by 2 (or pass 2Γ). The only non-vanishing elements are the following.

(a) Pair-transfer (“intra-pair”) block, for all p, r (including $p = r$):

$$\boxed{\Gamma_{r\bar{r}}^{p\bar{p}} = \Gamma_{\bar{r}r}^{\bar{p}p} = -\Gamma_{\bar{r}\bar{r}}^{pp} = -\Gamma_{rr}^{\bar{p}\bar{p}} = \frac{1}{2} D_{rp}} \quad (26)$$

For $r = p$ this is the on-pair element $\frac{1}{2} \langle \hat{n}_p \hat{n}_{\bar{p}} \rangle = \frac{1}{2} n_p$.

(b) Inter-pair density–density block, for $p \neq q$ and all $P \in \{p, \bar{p}\}$, $Q \in \{q, \bar{q}\}$ (16 elements per unordered pair $\{p, q\}$):

$$\boxed{\Gamma_{PQ}^{PQ} = \Gamma_{QP}^{QP} = -\Gamma_{QP}^{PQ} = -\Gamma_{PQ}^{QP} = \frac{1}{2} Q_{pq}} \quad (27)$$

i.e. Coulomb-type and exchange-type elements, both for same-Kramers ($P = p, Q = q$) and opposite-Kramers ($P = p, Q = \bar{q}$) combinations. Everything else is zero.

Trace (sum rule). The diagonal elements Γ_{PQ}^{PQ} come from block (a) with $r = p$ ($P, Q = p, \bar{p}$ and $\bar{p}, p$: $2 \times \frac{1}{2}n_p$) and from block (b) (4 Kramers combinations per ordered pair $p \neq q$: $4 \times \frac{1}{2}Q_{pq}$):

$$\boxed{\sum_{PQ} \Gamma_{PQ}^{PQ} = \sum_p n_p + 2 \sum_{p \neq q} Q_{pq} = \frac{N(N-1)}{2}} \quad (28)$$

This holds for *any* t, z , since $\langle \hat{N}(\hat{N}-1) \rangle = N(N-1) \langle 0|(1+Z)|0 \rangle = N(N-1)$. Check at $t = z = 0$: $n_{\text{occ}} + 2n_{\text{occ}}(n_{\text{occ}} - 1) = \frac{1}{2}N(N-1)$ with $N = 2n_{\text{occ}}$. Explicitly, with the formulas of Sec. 5.1,

$$\sum_p n_p + 2 \sum_{p \neq q} Q_{pq} = n_{\text{occ}} + 2 \sum_{i \neq j} (1 - x_i - x_j) + 4 \sum_{ia} (n_a - t_i^a z_a^i), \quad (29)$$

where the x -dependent terms cancel identically because $\sum_a n_a = \sum_i x_i$. Likewise the partial trace gives the 1-RDM: $\sum_Q \Gamma_{RQ}^{PQ} = \frac{N-1}{2} \gamma_R^P$.

Contracting (a) and (b) with the integrals reproduces Eq. (22): block (a) gives $\sum_{pr} \langle r\bar{r} | p\bar{p} \rangle D_{rp}$ (diagonal $\rightarrow V_p n_p$, off-diagonal $\rightarrow G_{rp} D_{rp}$); block (b) gives $\frac{1}{2} \sum_{p \neq q} W_{pq} Q_{pq}$.

Storage. The full spinor 2-RDM is determined by n_p , D_{pq} and Q_{pq} : $\mathcal{O}(n^2)$ real numbers. In a gradient/Hessian contraction the sparsity pattern above can be exploited directly (e.g. $\sum_{RST} \langle RS | PT \rangle \Gamma_{RS}^{QT}$ only needs $T \in \{Q, \bar{Q}\}$ -type index coincidences).

Relation to the spin-summed matrices of HBSS. Summing over Kramers labels as in HBSS Eq. (26) and restoring their normalisation ($\Gamma_{\text{HBSS}} = 2\Gamma$) gives $\Gamma_{pp}^{qq} = 2D_{pq}$, $\Gamma_{pq}^{pq} = 4Q_{pq}$, $\Gamma_{pq}^{qp} = -2Q_{pq}$ ($p \neq q$), i.e. HBSS Eqs. (A10). With spin-orbit coupling these Kramers-summed objects are *not* sufficient to evaluate E or the orbital gradient, because $\langle PQ | RS \rangle$ depends on the individual Kramers labels (e.g. $K_{p\bar{q}} \neq 0$). Use the spinor-resolved blocks (a), (b).

6 Remarks and checks

1. **Nonrelativistic limit.** See Sec. 7.
2. **Relativistic DOCI reference.** Eq. (6) is a hard-core-boson Hamiltonian; its exact diagonalisation in the $\binom{n}{n_{\text{occ}}}$ pair space is Kramers-restricted DOCI. This is trivial to code and gives an independent check of $E_{\text{pCCD}} \approx E_{\text{DOCI}}$ and of the RDMs (by direct evaluation of $\langle P_p^\dagger P_q \rangle$, $\langle N_p N_q \rangle$ on the biorthogonal pCCD vectors).
3. **Two-electron systems.** For a time-reversal-even two-electron state the pair function admits a Kramers-paired natural-spinor expansion, so oo-pCCD should equal FCI (as in the nonrelativistic case, HBSS Sec. IV); a good end-to-end test (e.g. He, H₂, or a Kramers-restricted active space of two electrons).
4. **Numerical verification.** The expressions in this note were checked by brute force: (i) Eqs. (16), (17) and all pair RDMs against explicit $\langle 0|(1+Z)e^{-T} O e^T |0 \rangle$ in the hard-core-boson space for random t, z, ε, G, W (errors $\sim 10^{-16}$; z equations against finite differences of $\mathcal{L}$, $\sim 10^{-10}$); (ii) the trace (28) for random t, z ; (iii) the reduction (6)–(10) and the spinor 2-RDM blocks (a), (b) against the full second-quantised spinor Hamiltonian (Jordan–Wigner, 6 spinors) built from random Kramers-restricted 2-component spinors with a time-reversal-even spin-orbit-like one-body term and a scalar two-body kernel.
5. **Orbital optimisation.** Because pairing is defined w.r.t. Kramers pairs, rotations within a Kramers pair are redundant; all inter-pair rotations (occ–occ, vir–vir, occ–vir) are non-redundant, as in HBSS. The frozen-pair extension (fpCCSD) would require the full spinor (non-pair) CC equations with $T_2^{(0)}$ fixed and is not covered here.

7 Recovery of the nonrelativistic limit

We show that for a spin-free Hamiltonian (nonrelativistic, or spin-free X2C/DC) and spin-pure Kramers pairs, every equation of this note reduces to the corresponding equation of HBSS.

7.1 Spin-pure Kramers pairs

Take h spin-free and a scalar two-body kernel, and build the pairs from real spatial orbitals ψ_p :

$$\phi_p = \psi_p \alpha, \quad \phi_{\bar{p}} = \mathcal{K} \phi_p = \psi_p^* \beta = \psi_p \beta, \quad (30)$$

using $\mathcal{K}\alpha = \beta$, $\mathcal{K}\beta = -\alpha$ for $\mathcal{K} = -i\sigma_y \mathcal{C}_0$. Then $P_p^\dagger = a_{p\alpha}^\dagger a_{p\beta}^\dagger = c_{p\uparrow}^\dagger c_{p\downarrow}^\dagger$, which is exactly HBSS Eq. (7) (same ordering, same phase), and the seniority operator is HBSS Eq. (1).

7.2 Integrals and the pair Hamiltonian

The spinor transition densities are

$$\rho_{pq} = \rho_{\bar{p}\bar{q}} = \psi_p \psi_q, \quad \rho_{p\bar{q}} = \psi_p \psi_q \alpha^\dagger \beta = 0. \quad (31)$$

Hence, in terms of spatial integrals $J_{pq} = (pp|qq)$ and $K_{pq} = (pq|qp)$,

$$J_{p\bar{q}} = J_{pq}, \quad K_{p\bar{q}} = 0, \quad V_p = \langle p\bar{p} || p\bar{p} \rangle = J_{pp}, \quad (32)$$

and Eqs. (7)–(9) (or (10)) become

$$\varepsilon_p = 2h_{pp} + J_{pp} = 2h_{pp} + v_{pp}^{pp}, \quad G_{pq} = K_{pq} = v_{qq}^{pp}, \quad W_{pq} = 4J_{pq} - 2K_{pq} = 2(2v_{pq}^{pq} - v_{qp}^{pq}), \quad (33)$$

with v_{rs}^{pq} the spatial Dirac-notation integrals of HBSS. The opposite-Kramers exchange disappears and only the same-spin exchange survives. The reference energy and Fock diagonal, Eqs. (11), (12), become

$$E_{\text{ref}} = \sum_i 2h_{ii} + \sum_{ij} (2J_{ij} - K_{ij}) = E_{\text{RHF}}, \quad f_{pp} = h_{pp} + \sum_j (2J_{pj} - K_{pj}) = f_p^p, \quad (34)$$

the RHF energy and the RHF Fock diagonal used in HBSS.

Spin-axis invariance. Spin purity with a common quantisation axis is not required. If each pair has its own spin direction, $\phi_p = \psi_p \chi_p$ with χ_p an arbitrary normalised two-spinor, then K_{pq} and $K_{p\bar{q}}$ are individually nonzero, but

$$K_{pq} + K_{p\bar{q}} = \left(|\langle \chi_p | \chi_q \rangle|^2 + |\langle \chi_p | \mathcal{K} \chi_q \rangle|^2 \right) K_{pq}^{\text{spatial}} = K_{pq}^{\text{spatial}}, \quad (35)$$

because $\{\chi_q, \mathcal{K}\chi_q\}$ is a complete orthonormal spin basis, and J_{pq} is unchanged. Since only the sum $K_{pq} + K_{p\bar{q}}$ enters G and W , Eq. (33) still holds. This is a special case of the invariance of $P_p^\dagger$, and hence of ε , G , W , under $SU(2)$ rotations within each Kramers pair.

7.3 Energy and amplitude equations

The pCCD energy becomes $E = E_{\text{RHF}} + \sum_{ia} v_{aa}^{ii} t_i^a$, HBSS Eq. (12a).

For the residual, insert Eq. (33) into the second form of Eq. (14):

$$\Delta_{ia} = 2(f_a^a - f_i^i) + v_{aa}^{aa} + v_{ii}^{ii} - 2(2v_{ia}^{ia} - v_{ai}^{ia}). \quad (36)$$

In HBSS Eq. (12b) the ladder sums $\sum_b v_{bb}^{aa} t_i^b$ and $\sum_j v_{ii}^{jj} t_j^a$ run over all b and j ; their diagonal terms $b = a$ and $j = i$ are $v_{aa}^{aa} t_i^a$ and $v_{ii}^{ii} t_i^a$. Splitting them off,

$$\sum_b v_{bb}^{aa} t_i^b + \sum_j v_{ii}^{jj} t_j^a = (v_{aa}^{aa} + v_{ii}^{ii}) t_i^a + \sum_{b \neq a} G_{ab} t_i^b + \sum_{j \neq i} G_{ji} t_j^a, \quad (37)$$

so the diagonal coefficient of t_i^a in HBSS Eq. (12b) is $2(f_a^a - f_i^i) + v_{aa}^{aa} + v_{ii}^{ii} - 2(2v_{ia}^{ia} - v_{ai}^{ia}) = \Delta_{ia}$. The remaining terms map one to one:

HBSS Eq. (12b)	Eq. (16)
v_{ii}^{aa}	G_{ia}
$-2(\sum_j v_{aa}^{jj} t_j^a + \sum_b v_{bb}^{ii} t_i^b) t_i^a$	$-2(\Sigma_a^{(v)} + \Sigma_i^{(o)}) t_i^a$
$+2v_{aa}^{ii} (t_i^a)^2$	$+2G_{ia} (t_i^a)^2$
$\sum_{jb} v_{bb}^{jj} t_j^a t_i^b$	$\sum_j t_j^a Y_{ji}$

Hence Eq. (16) = HBSS Eq. (12b). The same splitting of $\sum_b v_{aa}^{bb} z_b^i$ and $\sum_j v_{jj}^{ii} z_a^j$ gives Eq. (17) = HBSS Eq. (19), with $-2(2v_{ia}^{ia} - v_{ai}^{ia} - 2v_{aa}^{ii} t_i^a) z_a^i \rightarrow (-W_{ia} + 4G_{ia} t_i^a) z_a^i$ and $\sum_{jb} t_j^b (v_{bb}^{ii} z_a^j + v_{aa}^{jj} z_b^i) \rightarrow \sum_j z_a^j Y_{ij} + \sum_b z_b^i X_{ab}$.

7.4 Density matrices

The pair-level quantities n_p , D_{pq} , Q_{pq} depend only on t and z (the hard-core-boson algebra is the same), so they are unchanged. Identify $p \equiv p\alpha$, $\bar{p} \equiv p\beta$.

1-RDM. The spin-summed matrix of HBSS is $\gamma_q^p(\text{HBSS}) = \gamma_{q\alpha}^{p\alpha} + \gamma_{q\beta}^{p\beta} = 2n_p \delta_{pq}$, i.e. $\gamma_i^j = 2(1 - x_i) \delta_{ij}$ and $\gamma_a^b = 2x_a^a \delta_{ab}$: HBSS Eq. (A8).

2-RDM. With the Löwdin normalisation (24),

$$\Gamma_{rs}^{pq}(\text{HBSS}) = 2 \sum_{\sigma\sigma'} \Gamma_{r\sigma, s\sigma'}^{p\sigma, q\sigma'}. \quad (38)$$

From blocks (a) and (b):

- Γ_{pp}^{qq} : only $\sigma \neq \sigma'$ survive, giving $2 \times \frac{1}{2} D_{pq}$, hence $\Gamma_{pp}^{qq}(\text{HBSS}) = 2D_{pq}$. This yields $\Gamma_{ii}^{jj} = 2x_i^j$ ($i \neq j$), $\Gamma_{ii}^{ii} = 2n_i = 2(1 - x_i)$ [HBSS (A10a)], $\Gamma_{ii}^{aa} = 2D_{ia}$ [HBSS (A10b)], $\Gamma_{aa}^{ii} = 2z_a^i$ [HBSS (A10c)], $\Gamma_{aa}^{bb} = 2x_a^b$ [HBSS (A10d)].
- Γ_{pq}^{pq} , $p \neq q$: all four (σ, σ') survive, giving $4 \times \frac{1}{2} Q_{pq}$, hence $\Gamma_{pq}^{pq}(\text{HBSS}) = 4Q_{pq}$. This yields $\Gamma_{ij}^{ij} = 4(1 - x_i - x_j)$ [HBSS (A10e), $i \neq j$], $\Gamma_{ia}^{ia} = 4(x_a^a - t_i^a z_a^i)$ [HBSS (A10f)], $\Gamma_{ab}^{ab} = 0$ for $a \neq b$ [HBSS (A10g)]. For $p = q$ only $\sigma \neq \sigma'$ survive and one recovers the δ_{ij} and δ_{ab} terms of (A10e) and (A10g), both equal to $2n_p$.
- Γ_{pq}^{qp} , $p \neq q$: only $\sigma = \sigma'$ survive, giving $2 \times (-\frac{1}{2} Q_{pq})$, hence $\Gamma_{pq}^{qp}(\text{HBSS}) = -2Q_{pq} = -\frac{1}{2} \Gamma_{pq}^{pq}(\text{HBSS})$: HBSS (A10h).

The only difference is the overall factor 2 from the normalisation choice: $\sum_{pq} \Gamma_{pq}^{pq}(\text{HBSS}) = N(N - 1)$ versus $N(N - 1)/2$ here. The blocks (a) and (b) themselves reduce to the familiar $\alpha\beta$ -resolved pCCD 2-RDM, in which the opposite-spin exchange elements $\Gamma_{q\alpha, p\beta}^{p\alpha, q\beta}$ vanish.

7.5 Caveat: the orbital-optimisation space

The equations reduce exactly, but the stationary point reached by orbital optimisation need not be the HBSS one. Kramers-restricted (quaternion) rotations also include spin-mixing rotations, e.g. $\phi_p = \psi_1 \alpha + \psi_2 \beta$. The corresponding geminal $\phi_p(1) \mathcal{K} \phi_p(2)$ is time-reversal invariant but contains triplet components, so it is not an $\hat{S}^2$ eigenfunction. With a spin-free Hamiltonian:

- Starting from spin-pure pairs (e.g. localised RHF orbitals), the orbital gradient in the spin-mixing directions vanishes by symmetry, and the optimisation stays on the HBSS (singlet-paired) oo-pCCD solution.
- Starting from generic spinors, or following a negative Hessian eigenvalue in the spin-mixing block, the optimisation may reach a time-reversal-invariant but $\hat{S}^2$ -broken pairing with lower energy. This is the “non-singlet pairing” generalisation mentioned in the conclusions of HBSS.

For validation against HBSS or a nonrelativistic pCCD code, start from spin-pure pairs and check that the spin-mixing block of the orbital gradient vanishes. The lowest eigenvalue of the Hessian in that block indicates whether the singlet-paired solution is stable within the Kramers-restricted space.

Numerical check. With a spin-free one-body term and a scalar two-body kernel, the ε , G , W built from Eqs. (7)–(9) agree with Eq. (33) to machine precision, both for a common spin axis ($K_{p\bar{q}} = 0$) and for random per-pair spin axes ($K_{p\bar{q}} \neq 0$).